

वाणी सेतु · a bridge of voice

VaaniSetu

An AI-powered multilingual document reader built on AWS

Amazon Bedrock · Amazon Polly · Amazon Rekognition · AWS Lambda · API Gateway · S3 ·
DynamoDB · Amplify

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The problem

Everyday life runs on paper, but not everyone can read it comfortably



Low vision & elderly

Small print on prescriptions and bills is hard to read, and mistakes with doses or due dates have real consequences.



Limited literacy

Official notices and forms use formal, technical language that is hard to understand even when it can be read.



Language barrier

Migrants and students often receive documents in a language they do not read, such as a Hindi bill for a non-Hindi reader.

Our solution: VaaniSetu

Photograph a document and hear it explained in your language in about 10 seconds



Snap

Take a photo of a prescription, bill, notice or form



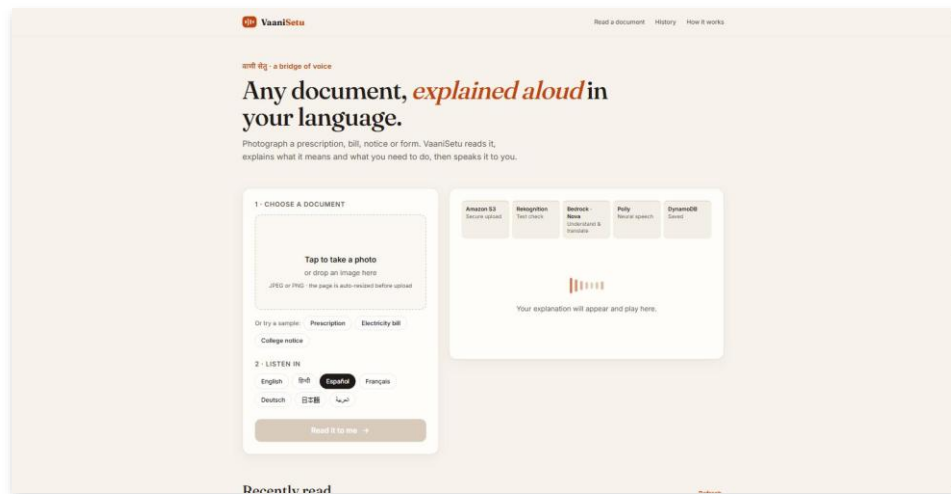
Understand

AI reads it and explains what it means and what to do



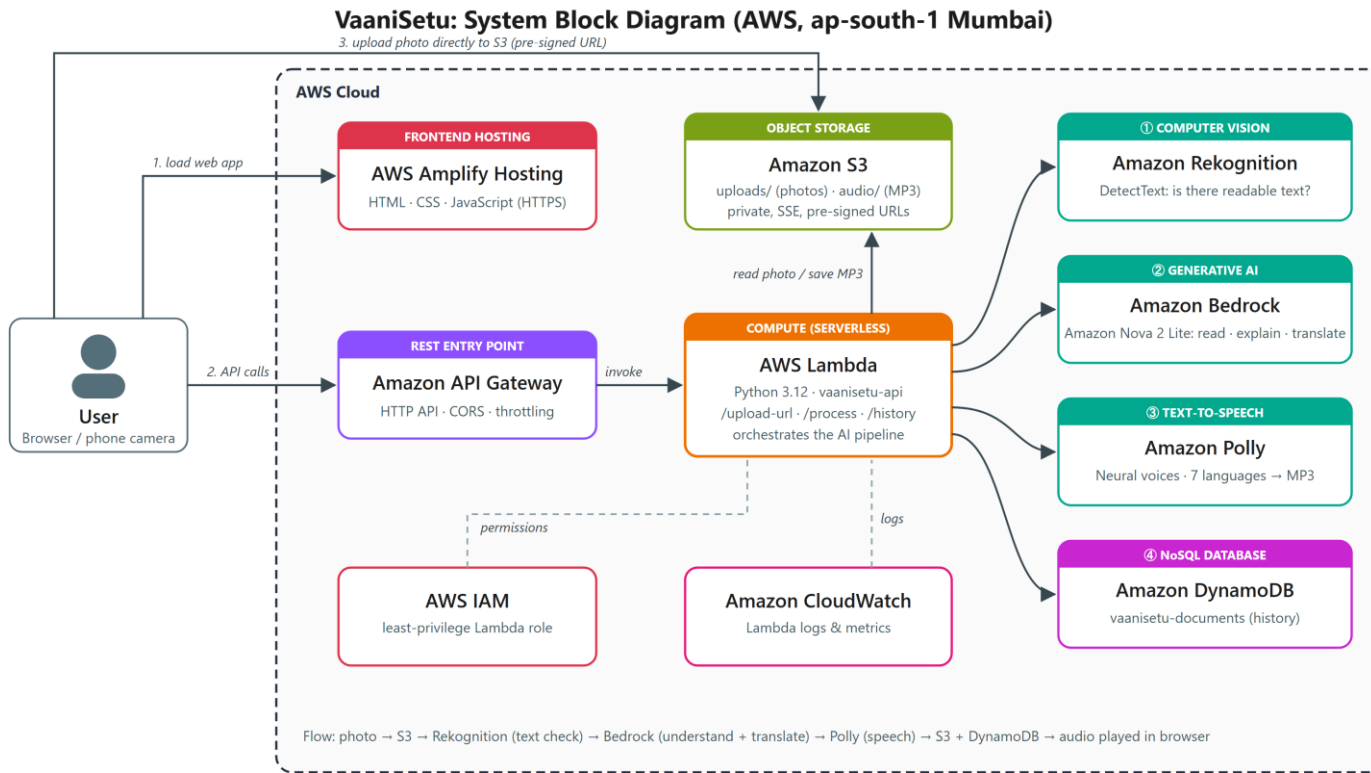
Listen

A natural neural voice reads the explanation aloud



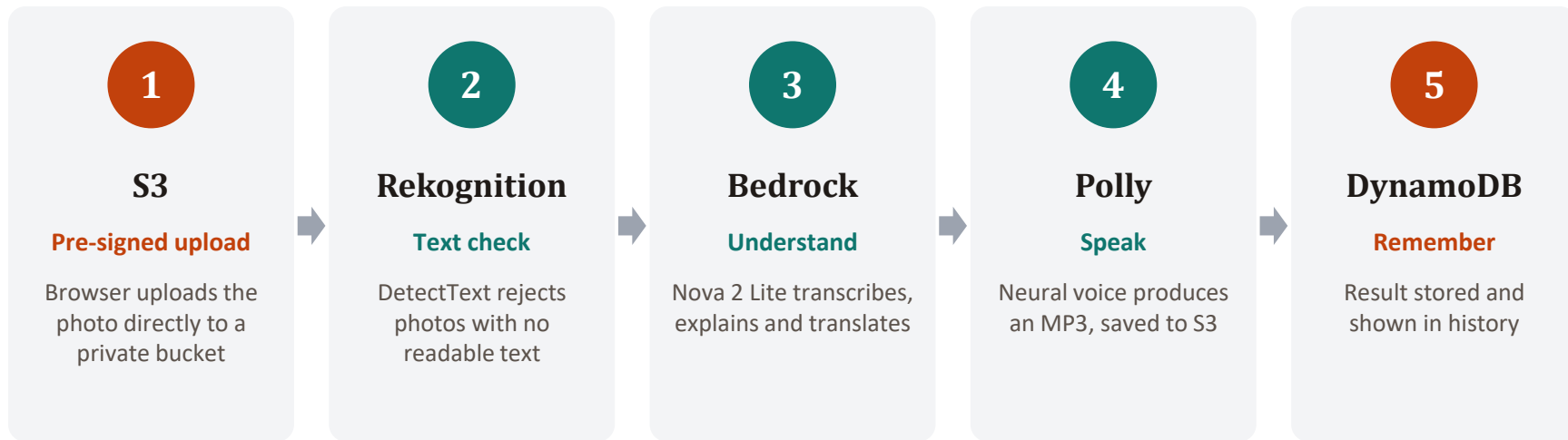
7 output languages · any input language · key points & action items · history

Block diagram



How a request flows

One photo, five AWS services, about 10 seconds



API Gateway (HTTP API) → AWS Lambda orchestrates every step · IAM least-privilege role · CloudWatch logs

Cloud services used

8 core AWS services plus IAM and CloudWatch, all serverless or fully managed



Amazon S3

Photos & MP3 audio,
private + encrypted



AWS Lambda

Python 3.12 backend
orchestrating the AI
pipeline



API Gateway

HTTP API: /upload-url,
/process, /history



Amazon Bedrock

Amazon Nova 2 Lite reads,
explains, translates



Rekognition

DetectText gate before the
AI model runs



Amazon Polly

Neural voices in 7
languages



DynamoDB

On-demand table for
document history



Amplify Hosting

HTTPS hosting + CDN for
the web app



AWS IAM

Least-privilege Lambda
execution role



CloudWatch

Logs & metrics for every
request

Development technologies



Python 3.12 + Boto3

Lambda backend calling S3, Rekognition, Bedrock Converse API, Polly and DynamoDB



HTML5

Semantic, accessible page structure with a skip link and ARIA roles



CSS3

Responsive grid, design tokens, animated pipeline tracker, reduced-motion support



JavaScript (ES2020)

Camera / drag-and-drop upload, Canvas resizing, Fetch API, audio playback



AWS CLI + Bash

One repeatable script provisions every resource



Git + GitHub

Version control with small, atomic commits



JSON

API payloads, structured AI output and IAM policies

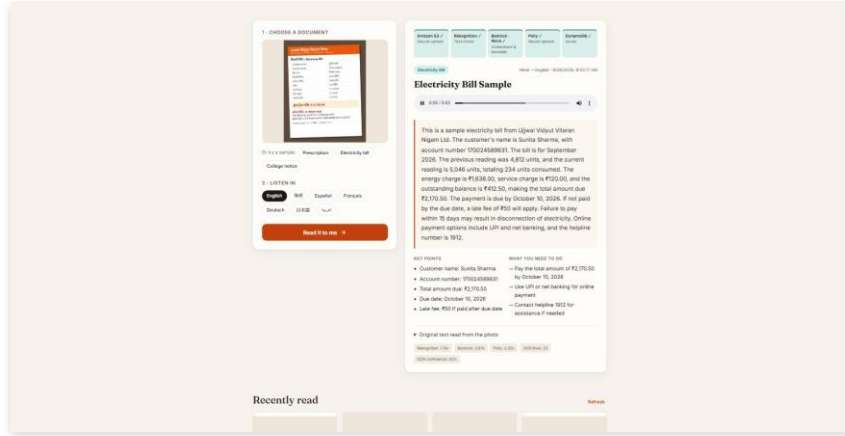
Demo: English notice → Hindi speech

The screenshot shows a web application interface for document-to-speech conversion. The interface is divided into several sections:

- 1- CHOOSE A DOCUMENT:** A section where a document titled "College notice" is selected. Below the document image, there are options to "Read it to me" and a "Read it to me" button.
- 2- LISTEN IN:** A section where the user can choose the language for the output speech. The "Hindi" option is selected.
- Document Preview:** A section showing the text of the document in Hindi. The text is: "यह नोटिस ऑनलाइन एडमिशन और टेक्नोलॉजी के उद्देश्य से एक सत्र के लिए है। सेमीनर की है ऑनलाइन एडमिशन 3 नवंबर 2026 से शुरू होगी। एडमिशन नॉटिस 8 अक्टूबर 2026 तक ऑनलाइन पर काम करना होगा। 12 अक्टूबर तक इसे से काम करने पर 500 रुपये का शुल्क लगेगा। 75% से कम उपस्थिति वाले सत्रों को हटा दिया नहीं जाएगा। हटा दिया 27 अक्टूबर 2026 से आधिकारिक तौर पर सत्रों को हटा दिया नहीं जाएगा। हटा दिया 31 अक्टूबर 2026 तक परीक्षाओं को जमा करने होगा। हर एडमिशन में ऑनलाइन एडमिशन काटें और हटा दिया सत्र से काम नहीं होगा। नोटिस पत्र और सत्र पत्रिका सत्रों से उपलब्ध है।"
- KEY POINTS:** A section listing key points in Hindi.
 - एडमिशन 3 नवंबर 2026 से शुरू होगा।
 - एडमिशन नॉटिस 8 अक्टूबर 2026 तक काम करना होगा।
 - 75% से कम उपस्थिति वाले सत्रों को हटा दिया नहीं जाएगा।
 - हटा दिया 27 अक्टूबर 2026 से आधिकारिक तौर पर सत्रों को हटा दिया नहीं जाएगा।
 - हटा दिया 31 अक्टूबर 2026 तक परीक्षाओं को जमा करने होगा।
 - हर एडमिशन में ऑनलाइन एडमिशन काटें और हटा दिया सत्र से काम नहीं होगा।
- WHAT YOU NEED TO DO:** A section listing what the user needs to do in Hindi.
 - एडमिशन नॉटिस 8 अक्टूबर 2026 तक काम करना होगा।
 - हटा दिया 27 अक्टूबर 2026 से आधिकारिक तौर पर सत्रों को हटा दिया नहीं जाएगा।
 - हर एडमिशन में ऑनलाइन एडमिशन काटें और हटा दिया सत्र से काम नहीं होगा।
- Original text read from the photo:** A section showing the original text read from the photo.
- Performance Metrics:** A section showing performance metrics: Recognition: 100%, Synthesis: 3.84s, Polly: 1.76s, OCR time: 10s, OCR confidence: 98.3%.
- Recently read:** A section showing a list of recently read documents.

- Document type detected: exam notice
- Summary written and spoken in Hindi by Polly (Kajal, neural)
- Key points and action items with deadlines
- All five pipeline stages complete
- OCR confidence 98.3%

Demo: Hindi bill → English, and history



A bill printed in Hindi is explained in English: amount due ₹2,170.50, due date 10 October 2026, late fee ₹50.



Every result is saved in DynamoDB. Tapping a card replays the explanation and audio.

The screenshot shows the AWS IAM console interface. The main content area is titled 'Routes' and contains a search bar and a 'Create' button. Below this, there is a table with columns for 'Name', 'Action', and 'Method'. The table is currently empty. The left sidebar shows the navigation menu with 'API Gateway' selected. The top navigation bar shows the user's name 'Amin' and the account ID '111111111111'.

The screenshot shows the Amazon S3 console interface. At the top, there's a navigation bar with 'Buckets', 'Objects', 'Properties', and a search bar. Below this, the 'audios/' bucket is selected. The 'Objects' tab is active, displaying a list of audio files. The table has columns: Name, Type, Last modified, Size, and Storage class. The objects are listed with their names, type (mp3), last modified dates, sizes (all 254.4 KB), and storage classes (all Standard). The console also shows a 'Files' section on the left with 'File systems' and 'Access management and analytics' options. The bottom of the console shows 'Storage management and insights' and 'Account and organization settings'.

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Performance at a glance

Measured per document during testing

~1 s

Recognition text check

~4 s

Bedrock understanding

~2 s

Polly neural speech

~10 s

Photo to speech, end to end

7

output languages

10

AWS services

₹0

cost while idle

Security & cost by design



Private storage

S3 public access blocked, SSE-S3 encryption at rest, uploads expire after 30 days



Pre-signed URLs

Browser gets 5-minute upload and 1-hour download links, never AWS keys



Least privilege

Lambda role may only use one bucket, one table, and the exact AI actions



Throttling

API Gateway limits 10 req/s (burst 20) to protect the budget



Cost gate

Rekognition blocks non-document photos before any AI charge



Serverless

No servers to patch; pay only per request

Challenges solved

Challenge

Textract, Translate & Comprehend blocked on the AWS Free plan



How it was solved

One multimodal Bedrock model reads, translates and summarises in a single call

CloudFront needs account verification for new accounts



Hosted the frontend on AWS Amplify Hosting (HTTPS + CDN)

Rekognition reads Latin script only



Used as a fast "is there text?" gate; Bedrock handles Hindi and other scripts

API Gateway's 29-second limit



Direct-to-S3 uploads and a pipeline that finishes in about 10 seconds

Conclusion & future scope

What was achieved

- A working, deployed serverless app that turns any document photo into spoken guidance
- 8 core AWS services + IAM and CloudWatch, integrated end to end
- Secure, cheap to run, and deployed with one script

What comes next

- Multi-page PDFs with SQS / Step Functions
- More Indian-language voices and a spoken Q&A mode
- Private per-user history with Amazon Cognito
- WhatsApp / IVR channel for feature phones

Thank you

Questions? · प्रश्न?